



Neet Test Hub Premium Test 15

Solution

Physics

1 - B	2 - D	3 - C	4 - B	5 - A	6 - B	7 - C	8 - D	9 - A	10 - C
11 - B	12 - D	13 - B	14 - C	15 - B	16 - A	17 - A	18 - D	19 - A	20 - B
21 - A	22 - B	23 - C	24 - A	25 - C	26 - D	27 - D	28 - B	29 - B	30 - C
31 - C	32 - A	33 - A	34 - A	35 - B	36 - A	37 - A	38 - A	39 - C	40 - C
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Chemistry

46 - C	47 - C	48 - D	49 - A	50 - A	51 - B	52 - D	53 - D	54 - D	55 - A
56 - D	57 - A	58 - B	59 - A	60 - C	61 - C	62 - C	63 - D	64 - B	65 - B
66 - B	67 - C	68 - B	69 - A	70 - A	71 - A	72 - A	73 - A	74 - B	75 - A
76 - A	77 - A	78 - A	79 - C	80 - D	81 - B	82 - C	83 - D	84 - A	85 - A
86 - C	87 - B	88 - A	89 - A	90 - C					

Biology - (Zoology)

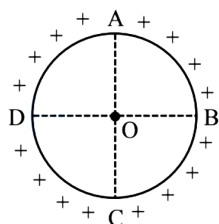
91 - A	92 - D	93 - A	94 - C	95 - C	96 - C	97 - B	98 - A	99 - B	100 - C
101 - C	102 - A	103 - D	104 - A	105 - C	106 - A	107 - A	108 - A	109 - D	110 - A
111 - D	112 - D	113 - A	114 - D	115 - B	116 - A	117 - C	118 - D	119 - D	120 - B
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131 - A	132 - D	133 - B	134 - A	135 - A					

Biology - (Botany)

136 - D	137 - B	138 - C	139 - A	140 - C	141 - B	142 - D	143 - D	144 - A	145 - A
146 - B	147 - B	148 - B	149 - A	150 - D	151 - A	152 - A	153 - C	154 - D	155 - C
156 - B	157 - B	158 - C	159 - D	160 - B	161 - A	162 - A	163 - D	164 - D	165 - C
166 - B	167 - B	168 - B	169 - D	170 - A	171 - D	172 - B	173 - C	174 - C	175 - B
176 - A	177 - C	178 - D	179 - B	180 - D					

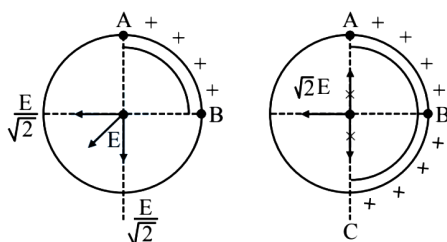
Physics

1. A metallic ring is uniformly charged as shown in figure. AC and BD are two mutually perpendicular diameters. Electric field due to arc AB to 'O' is 'E' is magnitude. What would be the magnitude of electric field at 'O' due to arc ABC?



- A) $2E$ **B) $\sqrt{2}E$** C) $E/2$ D) Zero

Solution : (Correct Answer: B)



2. The magnetic field in a plane electromagnetic wave is given by $B = 2 \times 10^{-7} \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)$. This electromagnetic wave is

- A) Visible light B) Infrared
C) Microwave **D) Radiowave**

Solution : (Correct Answer: D)

(d)

$$B = 2 \times 10^{-7} \sin(0.5 \times 10^{-3} x + 1.5 \times 10^{11} t)$$

$$f = \frac{c}{\lambda}, \quad k = 0.5 \times 10^{-3}$$

$$\omega = 1.5 \times 10^{11}$$

$$V = \frac{\omega}{k} = \frac{1.5 \times 10^{11}}{0.5 \times 10^{-3}} = 3 \times 10^{14}$$

$$\frac{\omega}{k} = f\lambda$$

$$k = \frac{2\pi}{\lambda}$$

$$\therefore \lambda = \frac{2\pi}{k} = \frac{2 \times 22}{7 \times 0.5 \times 10^{-3}} = 1.25 \times 10^4 \text{ m}$$

3. Maatch List I List II

Chose the correct answer from the options given below

List I		List II	
A.	$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$	I.	Gauss' law for electricity
B.	$\oint \vec{E} \cdot d\vec{l} = \frac{d\phi_B}{dt}$	II.	Gauss' law for magnetism
C.	$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$	III.	Faraday law
D.	$\oint \vec{B} \cdot d\vec{A} = 0$	IV.	Ampere - Maxwell law

- A) A - IV, B - I, C - III, D - II
B) A - II, B - III, C - I, D - IV
C) A - IV, B - III, C - I, D - II
D) A - I, B - II, C - III, D - IV

Solution : (Correct Answer: C)

Ampere - Maxwell law

$$\rightarrow \oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

$$\text{Faraday law} \rightarrow \oint \vec{E} \cdot d\vec{l} = \frac{d\phi_B}{dt}$$

$$\text{Gauss' law for electricity} \rightarrow \oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$\text{Gauss' law for magnetism} \rightarrow \oint \vec{B} \cdot d\vec{A} = 0$$

4. Identify the correct statements from the following descriptions of various properties of electromagnetic waves.

- A. In a plane electromagnetic wave electric field and magnetic field must be perpendicular to each other and direction of propagation of wave should be along electric field or magnetic field.
B. The energy in electromagnetic wave is divided equally between electric and magnetic fields.
C. Both electric field and magnetic field are parallel to each other and perpendicular to the direction of propagation of wave.
D. The electric field, magnetic field and direction of propagation of wave must be perpendicular to each other.

E. The ratio of amplitude of magnetic field to the amplitude of electric field is equal to speed of light.

Choose the most appropriate answer from the options given below:

- A) D only
 B) B and D only
 C) B, C and E only
 D) A, B and E only

Solution : (Correct Answer: B)

Second and fourth statements are correct.

5. A red LED emits light at 0.1 watt uniformly around it. The amplitude of the electric field of the light at a distance of 1 m from the diode is.... Vm^{-1}

- A) 2.45 B) 5.48
 C) 7.75 D) 9.73

Solution : (Correct Answer: A)

Using $U_{av} = \frac{1}{2} \epsilon_0 E^2$

But $U_{av} = \frac{P}{4\pi r^2 \times c}$

$$\frac{P}{4\pi r^2} = \frac{1}{2} \epsilon_0 E^2 \times c$$

$$E_0^2 = \frac{2P}{4\pi r^2 \epsilon_0 c} = \frac{2 \times 0.1 \times 9 \times 10^9}{1 \times 3 \times 10^8}$$

$$E_0 = \sqrt{6} = 2.45 \text{ V/m}$$

6. In which of the following, emission of electrons does not take place

- A) thermionic emission
 B) X-rays emission
 C) photoelectric emission
 D) secondary emission

Solution : (Correct Answer: B)

X-rays emission: They are due to transitions in the inner energy levels of the atom

Photoelectric emission : Emission of electrons from the metal surface on irradiation with radiation of surface frequency.

Secondary emission: When an electron strikes the surface of a metallic plate, it emits other electrons from the surface.

Thermionic emission: When a metal is heated to a high temperature, the free electron gain kinetic energy and escape from the surface of the metal.

7. The instantaneous voltages at three terminals marked X, Y and Z are given by

$$V_x = V_0 \sin \omega t \quad V_y = V_0 \sin \left(\omega t + \frac{2\pi}{3} \right) \quad \text{and} \quad V_z = V_0 \sin \left(\omega t + \frac{4\pi}{3} \right)$$

An ideal voltmeter is configured to read rms value of the potential difference between its terminals. It is connected between points X and Y and then between Y and Z. The reading(s) of the voltmeter will be

$$[A] V_{XY}^{rms} = V_0 \sqrt{\frac{3}{2}}$$

$$[B] V_{YZ}^{rms} = V_0 \sqrt{\frac{1}{2}}$$

$$[C] V_{XY}^{rms} = V_0$$

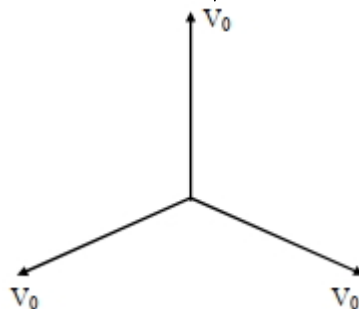
[D] independent of the choice of the two terminals

- A) A, C B) A, B C) A, D D) A, C, D

Solution : (Correct Answer: C)

$$V_{XYO} = V_{YZO} = V_{ZXO} = \sqrt{3} V_0$$

$$V_{XX}^{rms} = V_{YZ}^{rms} = \sqrt{\frac{3}{2}} V_0$$



8. A series LR circuit is connected to a voltage source with $V(t) = V_0 \sin \omega t$. After very large time, current $I(t)$ behaves as ($t_0 \gg \frac{L}{R}$)

- A) B) C) D)

Solution : (Correct Answer: D)

$$\text{Current } I = \frac{v}{\omega L} \left(\omega t - \frac{\pi}{2} \right)$$

9. If an AC main supply is given to be 220 V. The average emf during a positive half cycle will be.....V

- A) 198 B) 200 C) 240 D) $200\sqrt{2}$

Solution : (Correct Answer: A)

$$E_{rms} = 220V$$

$$\therefore E_{rms} = \frac{E_0}{\sqrt{2}} \Rightarrow E_0 = \sqrt{2} E_{rms}$$

Average e.m.f over half cycle

$$= \frac{2}{\pi} E_0 = 0.637 \times 1.41 \times 220 = 198.15 \text{ V}$$

10. The root mean square value of the alternating current is equal to

- A) Twice the peak value
 B) Half the peak value
 C) $\frac{1}{\sqrt{2}}$ times the peak value
 D) Equal to the peak value

Solution : (Correct Answer: C)

(c)

11. The r.m.s. voltage of domestic electricity supply is 220 volt . Electrical appliances should be designed to withstand an instantaneous voltage of.....V

A) 20 B) 310 C) 330 D) 440

Solution : (Correct Answer: B)

(b) Peak voltage = $\sqrt{2} \times 220 = 311 \text{ V}$

12. The magnetic flux linked with a coil of N turns of area of cross section A held with its plane parallel to the field B is

A) $\frac{NAB}{2}$ B) NAB C) $\frac{NAB}{4}$ D) zero

Solution : (Correct Answer: D)

Magnetic flux linked with a coil

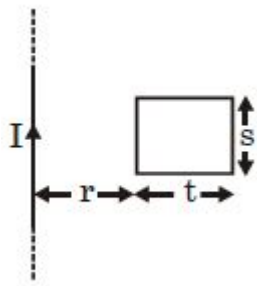
$$\phi = NBA \cos \theta$$

since the magnetic field B is parallel to the area A

i.e., $\theta = 90^\circ$.

$$\therefore \phi = 0$$

13. Flux passing through loop is proportional to



A) $\phi \propto r$
B) $\phi \propto s$
C) $\phi \propto t$
D) $\phi \propto r^{-1}$

Solution : (Correct Answer: B)

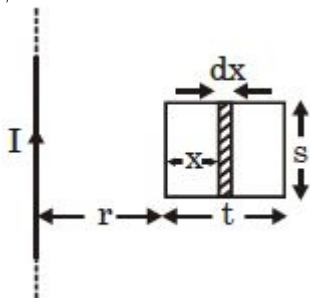
$$\int d\phi = \int_0^t \frac{\mu_0 I}{2\pi(r+x)} \cdot s \, dx$$

$$\phi = \frac{\mu_0 I s}{2\pi} \int_0^t \frac{dx}{r+x}$$

$$\phi = \frac{\mu_0 I s}{2\pi} \ln(r+x) \Big|_0^t$$

$$\phi = \frac{\mu_0 I s}{2\pi} \ln \left(\frac{r+t}{r} \right)$$

$$\phi \propto S$$



14. A closed planar wire loop of area A and arbitrary shape is placed in a uniform magnetic field of magnitude B , with its plane perpendicular to magnetic field. The resistance of the wire loop is R . The loop is now turned upside down by 180° so that its plane again becomes perpendicular to the magnetic field. The total charge that must have flowed through the wire ring in the process is

A) $< AB/R$ B) $= AB/R$
C) $= 2AB/R$ D) None

Solution : (Correct Answer: C)

$$\text{induced emf} = \frac{d(\text{flux})}{dt} = R \frac{dq}{dt}$$

$$\Rightarrow \Delta q = \frac{\Delta(\text{flux})}{R} = \frac{2BA}{R}$$

15. A conducting rod of length l is falling with a velocity v perpendicular to a uniform horizontal magnetic field B . The potential difference between its two ends will be

A) $2Blv$ B) Blv
C) $\frac{1}{2}Blv$ D) $B^2 l^2 v^2$

Solution : (Correct Answer: B)

(b) Force on a charge q in the rod = qvB

electric field inside the rod due to displacements of charges due to force by relative motion in magnetic field = $\frac{emf}{l}$

As the rod moves in constant velocity, net force of constituent charge q in the rod = 0

Therefore,

$$q \frac{emf}{l} = qvB$$

$$\Rightarrow emf = Blv$$

16. If the magnetic flux is expressed in weber, then magnetic induction can be expressed in

A) Weber/ m^2 B) Weber/ m
C) Weber - m D) Weber - m^2

Solution : (Correct Answer: A)

(a) Flux = $B \times A$; $\therefore B = \frac{\text{Flux}}{A} = \text{Weber}/m^2$

17. The magnetic potential due to a magnetic dipole at a point on its axis situated at a distance of 20 cm from its center is $1.5 \times 10^{-5} \text{ Tm}$. The magnetic moment of the dipole is _____ Am^2 . (Given : $\frac{\mu_0}{4\pi} = 10^{-7} \text{ TmA}^{-1}$)

A) 6 B) 5 C) 4 D) 12

Solution : (Correct Answer: A)

$$V = \frac{\mu_0}{4\pi} \frac{M}{r^2}$$

$$\Rightarrow 1.5 \times 10^{-5} = 10^{-7} \times \frac{M}{(20 \times 10^{-2})^2}$$

$$\Rightarrow M = \frac{1.5 \times 10^{-5} \times 20 \times 20 \times 10^{-4}}{10^{-7}}$$

$$M = 1.5 \times 4 = 6$$

18. The unit of magnetic moment is

- A) Wb/m B) $Wb.m^2$
C) $A.m$ D) $A.m^2$

Solution : (Correct Answer: D)

Magnetic dipole moment $M = \text{Current} \times \text{Area}$ So, its SI unit is $A.m^2$

19. The magnetic moment of a magnet of length 10 cm and pole strength 4.0 Am will be..... $A.m^2$

- A) 0.4 B) 1.6 C) 20 D) 8

Solution : (Correct Answer: A)

$$(a) M = mL = 4 \times 10 \times 10^{-2} = 0.4 A \times m^2$$

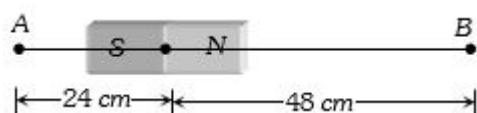
20. The magnetic compass is not useful for navigation near the magnetic poles because

- A) The magnetic field near the poles is zero
B) The magnetic field near the poles is almost vertical
C) At low temperature, the compass needle loses its magnetic properties
D) Neither of the above

Solution : (Correct Answer: B)

(b)

21. A bar magnet of length 3 cm has points A and B along its axis at distances of 24 cm and 48 cm on the opposite sides. Ratio of magnetic fields at these points will be



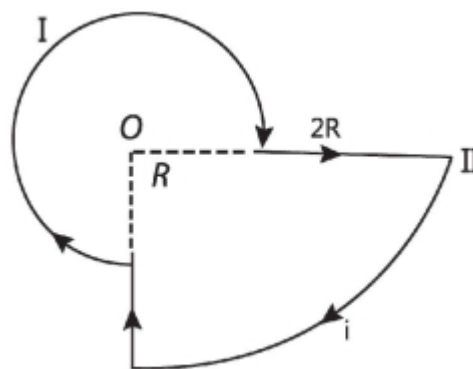
- A) 8 B) $1/2\sqrt{2}$ C) 3 D) 4

Solution : (Correct Answer: A)

(a) Both points A and B lying on the axis of the magnet and on axial position

$$B \propto \frac{1}{d^3} \Rightarrow \frac{B_A}{B_B} = \left(\frac{d_B}{d_A} \right)^3 = \left(\frac{48}{24} \right)^3 = 8$$

22. Consider the circular loop having current i and with central point O. The magnetic field at the central point O is



- A) $\frac{2\mu_0 i}{3\pi R}$ acting downward
B) $\frac{5\mu_0 i}{12R}$ acting downward
C) $\frac{6\mu_0 i}{11R}$ acting downward
D) $\frac{3\mu_0 i}{7R}$ acting upward

Solution : (Correct Answer: B)

(b)

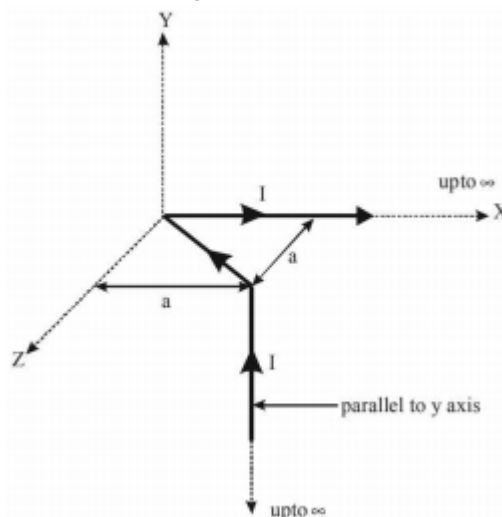
Magnetic field at the center due to the inner loop of radius $R = \frac{3}{4} \times \frac{\mu_0 I}{2R}$ downwards

Magnetic field at the center due to the outer loop of radius $3R = \frac{1}{4} \times \frac{\mu_0 I}{2(3R)}$ downwards

Magnetic field at center due to the two straight wire segments is zero.

$$\text{Hence net magnetic field at center of loop} = \frac{3\mu_0 I}{8R} + \frac{\mu_0 I}{24R} \text{ downwards} = \frac{5\mu_0 I}{12R} \text{ downwards}$$

23. The magnetic field at the origin due to the current flowing in the wire is -



- A) $-\frac{\mu_0 I}{8\pi a} (\hat{i} + \hat{k})$
B) $\frac{\mu_0 I}{2\pi a} (\hat{i} + \hat{k})$
C) $\frac{\mu_0 I}{8\pi a} (-\hat{i} + \hat{k})$
D) $\frac{\mu_0 I}{4\pi a\sqrt{2}} (\hat{i} - \hat{k})$

Solution : (Correct Answer: C)

$$B_{OD} = 0$$

$$B_{OB} = 0$$

$$B_{AB} = \frac{\mu_0}{4\pi a\sqrt{2}} [\cos 45^\circ (-\hat{i}) + \cos 45^\circ \hat{k}]$$

$$= \frac{\mu_0}{8\pi a} (-\hat{i} + \hat{k})$$

24. A particle is moving with velocity $\vec{v} = \hat{i} + 3\hat{j}$ and it produces an electric field at a point given by $\vec{E} = 2\hat{k}$. It will produce magnetic field at that point equal to (all quantities are in SI units)

- A) $\frac{6\hat{i}-2\hat{j}}{c^2}$
 B) $\frac{6\hat{i}+2\hat{j}}{c^2}$
 C) zero
 D) cannot be determined from the given data

Solution : (Correct Answer: A)

(a)

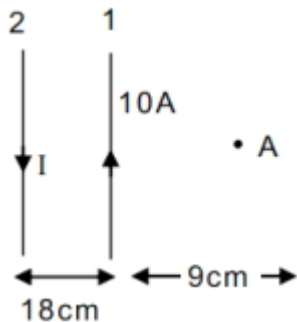
$$\vec{B} = \frac{\mu_0}{4\pi} q \frac{\vec{v} \times \vec{r}}{r^3} \text{ and } \vec{E} = \frac{1}{4\pi\epsilon_0} q \frac{\vec{r}}{r^3}$$

$$\mu_0 \epsilon_0 (\vec{v} \times \vec{E}) = \frac{\vec{v} \times \vec{E}}{c^2} = \frac{(\hat{i}+3\hat{j}) \times 2\hat{k}}{c^2}$$

$$= \frac{-2\hat{j}+6\hat{i}}{c^2} = \frac{6\hat{i}-2\hat{j}}{c^2}$$

$$\left[c = \frac{1}{\sqrt{\mu_0\epsilon_0}} \Rightarrow \mu_0 \epsilon_0 = \frac{1}{c^2} \right]$$

25. In figure two parallel infinitely long current carrying wires are shown. If resultant magnetic field at point A is zero. Then determine current I. (in A)



- A) 50 B) 15 C) 30 D) 25

Solution : (Correct Answer: C)

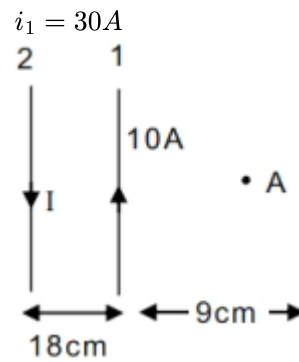
Consider the following figure

At point A

$$B_1 = B_2$$

$$\frac{\mu_0 i_1}{2\pi r_1} = \frac{\mu_0 i_2}{2\pi r_2}$$

$$\frac{i_1}{29} = \frac{10}{9}$$



26. An electron (charge q coulomb) enters a magnetic field of H weber/ m^2 with a velocity of v m/s in the same direction as that of the field the force on the electron is

- A) Hqv Newton's in the direction of the magnetic field
 B) Hqv dynes in the direction of the magnetic field
 C) Hqv Newton's at right angles to the direction of the magnetic field
 D) Zero

Solution : (Correct Answer: D)

(d) $\vec{F} = q(\vec{v} \times \vec{B}) = 0$ as $\vec{v}$ and $\vec{B}$ are parallel.

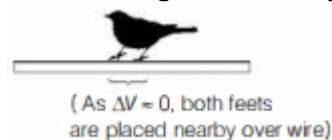
27. A bird sitting on a single high tension wire does not get electrocuted because

- A) the circuit is not complete
 B) resistance of the bird is too high
 C) capacitance of the bird is too small and the line frequency is too small
 D) Both a and b

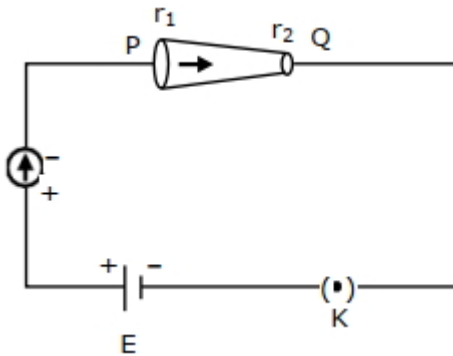
Solution : (Correct Answer: D)

(d)

When bird sits over a high tension wire, its both feet are nearly at same potential. So, no current flows through bird's body.



28. In the given figure, a battery of emf E is connected across a conductor PQ of length ' Y ' and different area of cross-sections having radii r_1 and r_2 ($r_2 < r_1$). Choose the correct option as one moves from P to Q :



- A) Electric field decreases.
B) Drift velocity of electron increases.
 C) Electron current decreases.
 D) All of these

Solution : (Correct Answer: B)

Current is constant in conductor

$$i = \text{constant}$$

$$\text{Resistance of element } dR = \frac{\rho dx}{\pi r^2}$$

$$dV = i dR = \frac{i \rho dx}{\pi r^2}$$

$$E = \frac{dV}{dx} = \frac{i \rho}{\pi r^2}$$

$$V_d = \frac{e E \tau}{m}$$

$$\therefore V_d \propto E$$

$$\rightarrow E \propto \frac{1}{r^2}$$

if r decreases, E will increase : V_d will increase

29. When 5 V potential difference is applied across a wire of length 0.1 m, the drift speed of electrons is $2.5 \times 10^{-4} \text{ ms}^{-1}$. If the electron density in the wire is $8 \times 10^{28} \text{ m}^{-3}$ the resistivity of the material is close to

- A) $1.6 \times 10^{-7} \Omega \text{m}$ **B) $1.6 \times 10^{-5} \Omega \text{m}$**
 C) $1.6 \times 10^{-3} \Omega \text{m}$ D) $1.6 \times 10^{-8} \Omega \text{m}$

Solution : (Correct Answer: B)

$$V = IR = (neAv_d) \rho \frac{\ell}{A}$$

$$\therefore \rho = \frac{V}{V_d n e \ell}$$

Here V = potential difference

ℓ = length of wire

n = no. of electrons per unit volume of conductor.

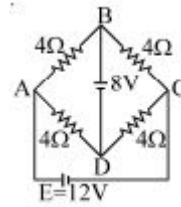
e = no. of electrons

Placing the value of above parameters we get resistivity

$$\rho = \frac{5}{8 \times 10^{28} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-4} \times 0.1}$$

$$= 1.6 \times 10^{-5} \Omega \text{m}$$

30. The given figure shows a network of resistances and a battery. Which of the two batteries is getting charged?



- A) 8 V battery B) 12 V battery
C) none D) can't be said

Solution : (Correct Answer: C)

31. A wire of diameter 0.02 meter contains 10^{28} free electrons per cubic meter. For an electrical current of 100 A, the drift velocity of the free electrons in the wire is nearly

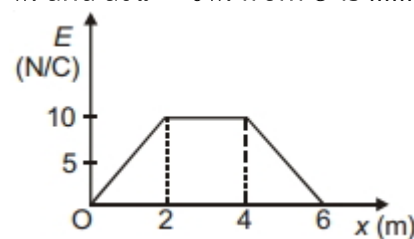
- A) $1 \times 10^{-19} \text{ m/s}$ B) $5 \times 10^{-10} \text{ m/s}$
C) $2 \times 10^{-4} \text{ m/s}$ D) $8 \times 10^3 \text{ m/s}$

Solution : (Correct Answer: C)

$$(c) \text{ By using } v_d = \frac{i}{neA} = \frac{100}{10^{28} \times 1.6 \times 10^{-19} \times \frac{\pi}{4} \times (0.02)^2}$$

$$= 2 \times 10^{-4} \text{ m/sec}$$

32. Figure shows the variation of electric field intensity E versus distance x . What is the potential difference between the points at $x = 2 \text{ m}$ and at $x = 6 \text{ m}$ from O is V



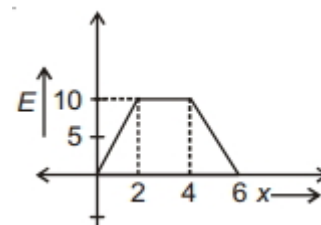
- A) 30** B) 60 C) 40 D) 80

Solution : (Correct Answer: A)

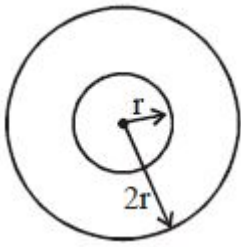
(a)

$$V_2 - V_6 = - \int E dx$$

$$V_2 - V_6 = (10)(2) + \frac{1}{2}(10)(2) = 30$$



33. If the potential of the inner shell is 10 V and that of the outer shell is 5 V , then potential at the centre will be.... V



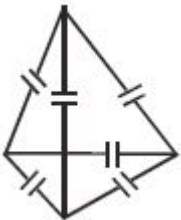
A) 10 B) 5 C) 15 D) 0

Solution : (Correct Answer: A)

$V_{\text{in}} = \frac{kQ_1}{a} + \frac{kQ_2}{2a} = 10\text{ V}$ (this will be constant for region from center to inner sphere surface)

$$V_{\text{out}} = \frac{kQ_1}{2a} + \frac{kQ_2}{2a} = 5\text{ V}$$

34. If the capacitance of each capacitor in the tetrahedron shown is C , then effective capacitance of the network across any two junctions is C



A) 2 B) 1 C) 0.5 D) 5

Solution : (Correct Answer: A)

One capacitor will be removed due to symmetry if we calculated then one of the capacitor will be removed and the remaining circuit will be

$$C_{\text{eq}} = C + \frac{C}{2} + \frac{C}{2} = 2C$$



35. In a region, if electric field is defined as $\vec{E} = (\hat{i} + 2\hat{j} + \hat{k})\text{ V/m}$, then the potential difference between two points $A(0,0,0)$ and $B(2,3,4)$ in that region, is V

A) 6 B) 12 C) 8 D) 9

Solution : (Correct Answer: B)

36. Equal charges are given to two spheres of different radii. The potential will

A) Be more on the smaller sphere
B) Be more on the bigger sphere
C) Be equal on both the spheres

- D) Depend on the nature of the materials of the spheres

Solution : (Correct Answer: A)

$$(a) V = \frac{kq}{R} \text{ i.e. } V \propto \frac{1}{R}$$

Potential on smaller sphere will be more.

37. An infinite plane sheet of charge having uniform surface charge density $+\sigma_5\text{ C/m}^2$ is placed on $x-y$ plane. Another infinitely long line charge having uniform linear charge density $+\lambda_e\text{ C/m}$ is placed at $z = 4\text{ m}$ plane and parallel to y -axis. If the magnitude values $|\sigma_s| = 2|\lambda_e|$ then at point $(0,0,2)$, the ratio of magnitudes of electric field values due to sheet charge to that of line charge is $\pi\sqrt{n} : 1$. The value of n is _____.

A) 16 B) 20 C) 23 D) 30

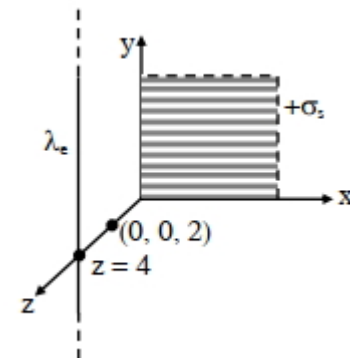
Solution : (Correct Answer: A)

$$\frac{E_S}{E_\ell} = \frac{\sigma}{2\epsilon_0} \times \frac{2\pi\epsilon_0 r}{\lambda}$$

$$= \frac{\pi \times \sigma r}{\lambda}$$

$$= \frac{\pi \times 2\lambda \times 2}{\lambda} = \frac{4\pi}{1}$$

$$\therefore n = 16$$



38. Find out the surface charge density at the intersection of point $x = 3\text{ m}$ plane and x -axis, in the region of uniform line charge of 8 nC/m lying along the z -axis in free space.

A) 0.424 nCm^{-2}
B) 47.88 C/m
C) 0.07 nCm^{-2}
D) 4.0 nCm^{-2}

Solution : (Correct Answer: A)

$$\frac{2K\lambda}{r} = \frac{\sigma}{\epsilon_0} \quad (x = 3\text{ m})$$

$$\sigma = 0.424 \times 10^{-9} \frac{\text{C}}{\text{m}^2}$$

39. Select the correct statement about electric charge

A) Charge can be converted into energy and energy can be converted into charge
B) Charge of a particle increases with increase in its velocity
C) Charge on a body is always integral multiple of a certain charge called charge of electron

D) Charge on a body is always positive or zero

Solution : (Correct Answer: C)

(c)

Quantization of charge.

40. Two identical tennis balls each having mass m and charge q are suspended from a fixed point by threads of length l . What is the equilibrium separation when each thread makes a small angle θ with the vertical?

- A) $x = \left(\frac{q^2 l}{2\pi\epsilon_0 mg} \right)^{1/2}$
 B) $x = \left(\frac{q^2 l^2}{2\pi\epsilon_0 m^2 g^2} \right)^{1/3}$
 C) $x = \left(\frac{q^2 l}{2\pi\epsilon_0 mg} \right)^{1/3}$
 D) $x = \left(\frac{q^2 l^2}{2\pi\epsilon_0 m^2 g} \right)^{1/3}$

Solution : (Correct Answer: C)

$$T \cos \theta = mg$$

$$T \sin \theta = \frac{kq^2}{x^2}$$

$$\tan \theta = \frac{kq^2}{x^2 mg}$$

$$\text{as } \tan \theta \approx \sin \theta \approx \frac{x}{2L}$$

$$\frac{x}{2L} = \frac{Kq^2}{x^2 mg}$$

$$x = \left(\frac{q^2 L}{2\pi\epsilon_0 mg} \right)^{1/3}$$

41. A neutral water molecule (H_2O) in its vapor state has an electric dipole moment of magnitude $6.4 \times 10^{-30} C - m$. How far apart are the molecules centres of positive and negative charge

- A) $4 m$ B) $4 mm$
 C) $4 pm$ D) $4 \mu m$

Solution : (Correct Answer: C)

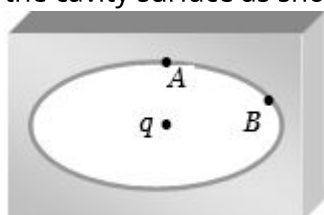
(c) There are 10 electrons and 10 protons in a neutral water molecule.

So its dipole moment is $p = q(2l) = 10e(2l)$

Hence length of the dipole i.e. distance between centres of positive and negative charges is

$$2l = \frac{p}{10e} = \frac{6.4 \times 10^{-30}}{10 \times 1.6 \times 10^{-19}} = 4 \times 10^{-12} m = 4 pm$$

42. An ellipsoidal cavity is carved within a perfect conductor. A positive charge q is placed at the centre of the cavity. The points A and B are on the cavity surface as shown in the figure. Then



- A) Electric field near A in the cavity = Electric field near B in the cavity

B) Total electric field flux through the surface of the cavity is q/ϵ_0

C) Potential at A = Potential at B

D) Both (b) and (c)

Solution : (Correct Answer: D)

(d) Under electrostatic condition, all points lying on the conductor are in same potential. Therefore, potential at A = potential at B .

From Gauss's theorem, total flux through the surface of the cavity will be q/ϵ_0 .

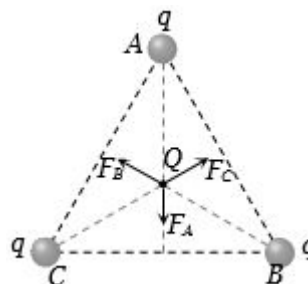
Note : Instead of an elliptical cavity, if it would have been a spherical cavity then options (a) and (b) were also correct.

43. Three charges each of magnitude q are placed at the corners of an equilateral triangle, the electrostatic force on the charge placed at the center is (each side of triangle is L)

- A) Zero
 B) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{L^2}$
 C) $\frac{1}{4\pi\epsilon_0} \frac{3q^2}{L^2}$
 D) $\frac{1}{12\pi\epsilon_0} \frac{q^2}{L^2}$

Solution : (Correct Answer: A)

(a) In the following figure since $|\vec{F}_A| = |\vec{F}_B| = |\vec{F}_C|$ and they are equally inclined with each other, so their resultant will be zero.



44. One metallic sphere A is given positive charge whereas another identical metallic sphere B of exactly same mass as of A is given equal amount of negative charge. Then

- A) Mass of A and mass of B still remain equal
 B) Mass of A increases
 C) Mass of B decreases
 D) Mass of B increases

Solution : (Correct Answer: D)

(d) Negative charge means excess of electron which increases the mass of sphere B .

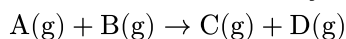
45. Charge on α - particle is

- A) $4.8 \times 10^{-19} C$ B) $1.6 \times 10^{-19} C$
 C) $3.2 \times 10^{-19} C$ D) $6.4 \times 10^{-19} C$

Solution : (Correct Answer: C)

(c) By using $Q = ne \Rightarrow Q = +2e = +3.2 \times 10^{-19} C$

46. Consider an elementary reaction



If the volume of reaction mixture is suddenly reduced to $\frac{1}{3}$ of its initial volume, the reaction rate will become 'x' times of the original reaction rate. The value of x is :

- A) 3 B) $\frac{1}{9}$ C) 9 D) $\frac{1}{3}$

Solution : (Correct Answer: C)

Since, the reaction is elementary

$$\text{Rate} = k[A]^1[B]^1$$

When V is reduced to $\frac{1}{3}$ V, concentration will be tripled.

Hence, rate = $9 \times (\text{rate})_{\text{initial}}$

$$x = 9$$

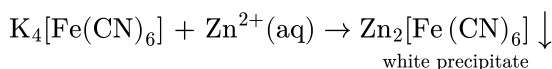
47. $K_4[Fe(CN)_6] + M^{x+}(aq) \rightarrow M_4[Fe(CN)_6]_x \downarrow$
Coloured precipitate

Which of the following cation does not respond to the above reaction?

- A) $Cu^{2+}(aq)$ B) $Fe^{3+}(aq)$
 C) $Zn^{2+}(aq)$ D) None of these

Solution : (Correct Answer: C)

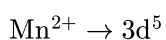
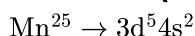
Here only Zn^{2+} do not form coloured precipitate here



48. Calculate the number of unpaired electrons in $[Mn(H_2O)_6]^{2+}$, Considering H_2O as a weak field ligand (At. No. of Mn = 25)

- A) Two B) Three C) Four D) Five

Solution : (Correct Answer: D)



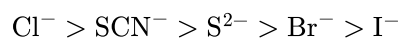
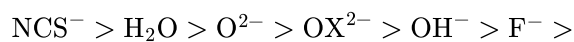
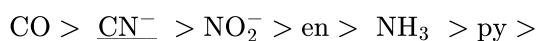
In presence of weak ligand field, there will be no pairing of electrons. So, it will form a high spin complex. i.e. the number of unpaired electrons = 5.

49. In which of the following octahedral complexes of Co (At. No. 27), will the magnitude of Δ_0 be the highest?

- A) $[Co(CN)_6]^{3-}$ B) $[Co(C_2O_4)_3]^{3-}$
 C) $[Co(H_2O)_6]^{3+}$ D) $[Co(NH_3)_6]^{3+}$

Solution : (Correct Answer: A)

CFSE (crystal field splitting energy) for octahedral complex, Δ_0 depends on the strength of ligand. Spectrochemically it has been found that the strength of splitting is as follows



Therefore, magnitude of Δ_0 will be highest in case of $[Co(CN)_6]^{3-}$.

50. $[Fe(NO_2)_3Cl_3]$ and $[Fe(O-NO)_3Cl_3]$ are

- A) linkage isomers
 B) geometrical isomers
 C) optical isomers
 D) hydrate isomers

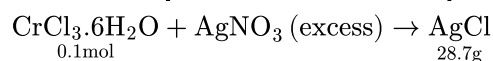
Solution : (Correct Answer: A)

$[Fe(NO_2)_3Cl_3]$ and $[Fe(O-NO)_3Cl_3]$ shows linkage isomerism.

51. For the complex with molecular formula $CrCl_3 \cdot 6H_2O$ 0.1 mole of the complex when treated with excess of $AgNO_3$ gave 28.7 g of white precipitate of $AgCl$ (Molar mass = 143.5 g). The formula of the complex would be

- A) $[Cr(H_2O)_6]Cl_3$ B) $[CrCl(H_2O)_5]Cl_2 \cdot H_2O$
 C) $[CrCl_2(H_2O)_4]Cl \cdot 2H_2O$ D) $[Cr(H_2O)_3Cl_3] \cdot 3H_2O$

Solution : (Correct Answer: B)



$$\text{No. of moles of } AgCl = \frac{28.7}{143.5} \approx 0.2 \text{ mol}$$

52. Potassium dichromate is used -

- A) As a reducing agent
 B) In electroplating
 C) As an insecticide
 D) as an oxidizing agent it oxidises ferrous ions into ferric ions in acidic media

Solution : (Correct Answer: D)

Potassium dichromate ($K_2Cr_2O_7$), is a common inorganic chemical reagent, most commonly used as an oxidizing agent in various laboratory and industrial applications.

Ferrous iron is oxidized to ferric iron by potassium dichromate in acid solution. The completion of oxidation reaction marked by the appearance of blue violet color of diphenylamine, which is used as an internal indicator.

53. The radius of La^{3+} (Atomic number of La=57) is 1.06 Å. Which one of the following given values will be closest to the radius of Lu^{3+} (Atomic number of Lu=71)?

- A) 1.60 Å B) 1.40 Å C) 1.06 Å D) 0.85 Å

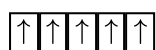
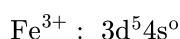
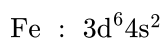
Solution : (Correct Answer: D)

Due to lanthanide contraction, the ionic radii of Ln^{3+} (lanthanide ions) decreases from La^{3+} to Lu^{3+} . Thus the lowest value (here $0.85 \text{ \AA}$) is the ionic radius of Lu^{3+} .

54. Calculate the value of magnetic moment for ${}_{26}\text{Fe}^{3+}$.

A) 1.73 BM B) 3.87 BM
C) 4.90 BM D) 5.92 BM

Solution : (Correct Answer: D)



$$n = 5$$

The formula of magnetic moment (μ):

$$\sqrt{n(n+2)} \text{ BM}$$

Whereas:- n = Number of unpaired electron

BM = Bohr Magnetron

$$\mu = \sqrt{35} \text{ BM}$$

$$= 5.92 \text{ BM}$$

55. The lanthanide contraction is responsible for the fact that

A) Zr and Hf have same radius
B) Zr and Zn have the same oxidation state
C) Zr and Y have same radius
D) Zr and Nb have similar oxidation state

Solution : (Correct Answer: A)

Due to lanthanide contraction effective nuclear charge of 5d series metals increases and size decreases.

56. Although Zirconium belongs to 4d and hafnium belongs to 5d transition series, they show similar physical and chemical properties because both

A) belong to d-block
B) have same number of electrons
C) belongs to the same group of the periodic table
D) have similar atomic radius

Solution : (Correct Answer: D)

This is because the atomic radii of 4d and 5d transition elements are nearly same. This similarity in size ($\text{Zr} = 160$, $\text{Hf} = 159 \text{ pm}$) is due to lanthanoid contraction. Because of this contraction the radius of Hf becomes nearly equal to that of Zr and as the

properties of an element depends on its atomic radii they show similar properties.

57. The coordination number of Cu in complex $[\text{Cu}(\text{H}_2\text{O})_4]^{++}$ is

A) 4 B) 3 C) 2 D) 1

Solution : (Correct Answer: A)

(a) The co-ordination no. = no. of ligands attached.

58. Co-ordination number of Fe in the complexes $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{FeCl}_4]^-$ would be respectively

A) 2, 3, 3 B) 6, 6, 4
C) 6, 3, 3 D) 6, 4, 6

Solution : (Correct Answer: B)

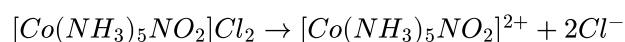
Co-ordination number is equal to total number of ligands in a complex.

59. A co-ordination complex compound of cobalt has the molecular formulae containing five ammonia molecules, one nitro group and two chlorine atoms for one cobalt atom. One mole of this compound produces three mole ions in an aqueous solution on reacting with excess of AgNO_3 , AgCl precipitate. The ionic formula for this complex would be:

A) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$
B) $[\text{Co}(\text{NH}_3)_5\text{Cl}][\text{Cl}(\text{NO}_2)]$
C) $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)\text{Cl}][(\text{NH}_3)\text{Cl}]$
D) $[\text{Co}(\text{NH}_3)_5][(\text{NO}_2)_2\text{Cl}_2]$

Solution : (Correct Answer: A)

(a) The most probable complex which gives three moles ions in aqueous solution may be $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ because it gives two chlorine atoms on ionisation.



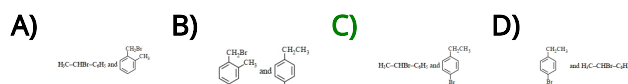
60. Cobalt (III) chloride forms several octahedral complexes with ammonia. Which of the following will not give test for chloride ions with silver nitrate at 25°C ?

A) $\text{CoCl}_3 \cdot 5\text{NH}_3$ B) $\text{CoCl}_3 \cdot 6\text{NH}_3$
C) $\text{CoCl}_3 \cdot 3\text{NH}_3$ D) $\text{CoCl}_3 \cdot 4\text{NH}_3$

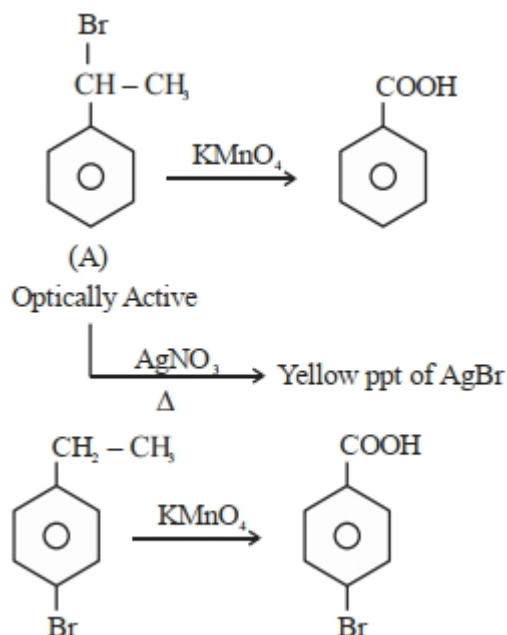
Solution : (Correct Answer: C)

Octahedral complex means it contains '6' coordinate bond. In $\text{CoCl}_3 \cdot 3\text{NH}_3$, Cl cannot be outside from coordination sphere

61. Two isomers (A) and (B) with Molar mass 184 g/mol and elemental composition C, 52.2%; H, 4.9% and Br 42.9% gave benzoic acid and *p*-bromobenzoic acid, respectively on oxidation with $KMnO_4$. Isomer 'A' is optically active and gives a pale yellow precipitate when warmed with alcoholic $AgNO_3$. Isomer 'A' and 'B' are, respectively



Solution : (Correct Answer: C)



62. Which *d*-block element is expected to have highest enthalpy of atomisation, among the following?

- A) Sc B) Mn
C) V, Zn D) None of these

Solution : (Correct Answer: C)

Enthalpy of atomisation is the change in enthalpy when one mole of bonds are completely broken to obtain atoms in the gaseous phase. Transition elements have high effective nuclear charge and a large number of valence electrons. Therefore, they form very strong metallic bonds as a result, the enthalpy of atomisation of transition metal is high. Due to the absence of unpaired electrons, the inter-atomic electronic bonding is the weakest in Zn and as a result, it has the least enthalpy of atomisation due to presence of 3 unpaired electrons. Vanadium have the highest atomisation enthalpy.

63. Iron exhibits +2 and +3 oxidation states. Which of the following statements about iron is incorrect?

- A) Ferrous oxide is more basic in nature than the ferric oxide.
B) Ferrous compounds are relatively more ionic than the corresponding ferric compounds.
C) Ferrous compounds are less volatile than the corresponding ferric compounds.
D) Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.

Solution : (Correct Answer: D)

Fe^{3+} is easily hydrolysed than Fe^{2+} due to more positive charge.

64. The highest magnetic moment is shown by the transition metal ion with the outer electronic configuration

- A) $3d^2$ B) $3d^5$
C) $3d^7$ D) $3d^9$

Solution : (Correct Answer: B)

(b) $3d^5$ as this configuration corresponds to maximum number of unpaired electrons.

65. What is the name of element with atomic number 105

- A) Kurchatovium B) Dubnium C) Nobelium D) Holmium

Solution : (Correct Answer: B)

(b) The element with atomic no. 105 is dubnium. In IUPAC nomenclature, it is known as un-nil pentin.

66. Which of the following represents the electronic configuration of a transition element

- A) $1s^2, 2s^2p^6 \dots ns^2p^3$
B) $1s^2, 2s^2p^6 \dots ns^2p^6d^3, (n+1)s^2$
C) $1s^2, 2s^2p^6 \dots ns^2p^6d^{10}, (n+1)s^2p^1$
D) $1s^2, 2s^2p^6 \dots ns^2p^6$

Solution : (Correct Answer: B)

(b) $1s^2, 2s^2p^6 \dots ns^2p^6d^3, (n+1)s^2$

as last electron enters *d*-subshell.

67. Rate constant varies with temperature by the equation $\log_{10} K = 5 - 2000/T$. We can conclude that ($R = 8.314 J mol^{-1} K^{-1}$)

- A) Pre exponential factor *A* is 5
B) E_a is 4 kcal/mol
C) Pre exponential factor *A* is 10^5
D) E_a is 19.212 kcal/mol

Solution : (Correct Answer: C)

68. The rate of reaction $A + 2B \rightarrow 3C$ becomes 72 times when concentration of *A* is tripled and concentration of *B* is doubled then the order of reaction with respect to *A* and *B* respectively is

A) 1, 2 B) 2, 3 C) 3, 2 D) 2, 2

Solution : (Correct Answer: B)

$r = k[A]^\alpha[B]^\beta \dots$ (i) $72r = k[3A]^\alpha[2B]^\beta \dots$ (ii) Dividing eqn. (i) by eqn (ii) $72 = 3^\alpha 2^\beta$ By substituting the given values of $\alpha = 2, \beta = 3$ we have $3^\alpha \times 2^\beta = 3^2 \times 2^3 = 9 \times 8 = 72$

69. NO_2 required for a reaction is produced by the decomposition of N_2O_5 in CCl_4 as per the equation $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$ The initial concentration of N_2O_5 is 3.00 mol L^{-1} and it is 2.75 mol L^{-1} after 30 minutes. The rate of formation of NO_2 is $\text{mol L}^{-1} \text{ min}^{-1}$

A) 1.667×10^{-2} B) 4.167×10^{-3}
C) 8.333×10^{-3} D) 2.083×10^{-3}

Solution : (Correct Answer: A)



$$t = 0 \quad 3.0 \text{ M}$$

$$t = 30 \quad 2.75 \text{ M}$$

$$\frac{-\Delta[N_2O_5]}{\Delta t} = \frac{0.25}{30}$$

$$\frac{1}{2} \times \frac{-\Delta[N_2O_5]}{\Delta t} = \frac{1}{4} \times \frac{-\Delta[NO_2]}{\Delta t}$$

$$\frac{\Delta[NO_2]}{\Delta t} = \frac{0.25}{30} \times 2 = 1.66 \times 10^{-2} \text{ M/min}$$

70. In a first order reaction the $a/(a-x)$ was found to be 8 after 10 minute. The rate constant is

A) $\frac{(2.303 \times 3 \log 2)}{10}$
B) $\frac{(2.303 \times 2 \log 3)}{10}$
C) $10 \times 2.303 \times 2 \log 3$
D) $10 \times 2.303 \times 3 \log 2$

Solution : (Correct Answer: A)

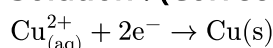
$$kt = 2.303 \log_{10} \frac{a}{a-x}$$

$$k \times 10 = 2.303 \log_{10} 8$$

71. $Cu^{2+} + 2e^- \rightarrow Cu$. On increasing $[Cu^{2+}]$ concentration, electrode potential

A) Increases
B) Decreases
C) Will remain unchanged
D) First increases, then decreases

Solution : (Correct Answer: A)

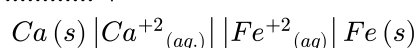


$$E_{Cu^{2+}/Cu} = E^0_{Cu^{2+}/Cu} - \frac{0.06}{2} \log \left| \frac{1}{Cu^{2+}} \right|$$

$$= E^0_{Cu^{2+}/Cu} + \frac{0.06}{2} \log |Cu^{2+}|$$

$|Cu^{2+}|$ increases, $E_{Cu^{2+}/Cu}$ also increases

72. Calculate E^0_{cell} of following galvanic cell at 298 K ;
..... V



$$E^0_{Ca^{+2}/Ca} = -2.87 \text{ V} ; E^0_{Fe/Fe^{+2}} = 0.41 \text{ V}$$

A) 2.46 B) -2.46 C) 3.28 D) -3.28

Solution : (Correct Answer: A)

73. Electrolysis of molten anhydrous calcium chloride produces

A) Calcium B) Phosphorus C) Sulphur D) Sodium

Solution : (Correct Answer: A)

(a) Calcium is produced when molten anhydrous calcium chloride is electrolysed.

74. Electrolysis rules of Faraday's states that mass deposited on electrode is proportional to

A) $m \propto I^2$
B) $m \propto Q$
C) $m \propto Q^2$
D) None of these

Solution : (Correct Answer: B)

(b) It is Faraday's law.

75. Electrolysis involves oxidation and reduction respectively at

A) Anode and cathode
B) Cathode and anode
C) At both the electrodes
D) None of the above

Solution : (Correct Answer: A)

(a) In electrolysis process oxidation occurs at anode and reduction occurs at cathode.

76. Mole fraction of O_2 in O_2 and O_3 mixture is 0.25. Hence percentage concentration ($\frac{w}{W}\%$) O_2 in mixture is

A) 18.18 B) 1.81 C) 81.82 D) 8.18

Solution : (Correct Answer: A)

$$X_{O_2} = \frac{n_{O_2}}{n_{O_2} + n_{O_3}} = 0.25$$

$$X_{O_3} = \frac{n_{O_3}}{n_{O_2} + n_{O_3}} = 0.75$$

$$\% \frac{w}{w} \text{ of } O_2 = \frac{\text{wt of } O_2}{\text{wt of } O_2 + \text{wt of } O_3} \times 100\%$$

$$\frac{w_{O_2}}{w_{O_2} + w_{O_3}} \times 100 = \frac{32}{32 + 144} \times 100 = \frac{32}{176} \times 100 = 18.18$$

77. On mixing, heptane and octane form an ideal solution. At 373 K, the vapour pressures of the two liquid components (heptane and octane) are 105 kPa and 45 kPa respectively. Vapour pressure of the solution obtained by mixing 25.0 g of heptane and 35 g of octane will be kPa.
(molar mass of heptane = 100 g mol⁻¹ and of octane = 114 g mol⁻¹)

A) 72.0 B) 36.1 C) 96.2 D) 144.5

Solution : (Correct Answer: A)

$$\begin{aligned}
 P_{\text{Total}} &= P_A^\circ x_A + P_B^\circ x_B \\
 &= P_{\text{Heptane}}^\circ x_{\text{heptane}} + P_{\text{Octane}}^\circ x_{\text{Octane}} \\
 &= 105 \times \frac{25/100}{\frac{2}{100} + \frac{33}{114}} + 45 \times \frac{35/114}{\frac{25}{100} + \frac{33}{114}} \\
 &= 105 \times \frac{0.25}{0.25+0.3} + 45 \times \frac{0.3}{0.25+0.3} \\
 &= \frac{105 \times 0.25}{0.55} + \frac{45 \times 0.3}{0.55} \\
 &= \frac{26.25 + 13.5}{0.55} = 72 \text{ kPa}
 \end{aligned}$$

78. In which case Raoult's law is not applicable

- A) 1M NaCl B) 1M urea
C) 1M glucose D) 1M sucrose

Solution : (Correct Answer: A)

Raoult's law is not applicable if the total number of particles of solute changes in the solution due to association or dissociation.

Urea, glucose and sucrose are the non-electrolytes. They cannot dissociate into ions.

NaCl dissociates into Na^+ ions and Cl^- ions.

Hence, option A is correct.

79. In an experiment, 1 g of a non-volatile solute was dissolved in 100 g of acetone (mol. mass = 58) at 298 K. The vapour pressure of the solution was found to be 192.5 mm Hg. The molecular weight of the solute is (vapour pressure of acetone = 195 mm Hg)

- A) 25.24 B) 35.24
C) 45.24 D) 55.24

Solution : (Correct Answer: C)

It's Obvious.

80. A solution has a 1 : 4 mole ratio of pentane to hexane. The vapour pressure of the pure hydrocarbons at 20°C are 440 mm Hg for pentane and 120 mm Hg for hexane. The mole fraction of pentane in the vapour phase would be

- A) 0.549 B) 0.2 C) 0.786 D) 0.478

Solution : (Correct Answer: D)

$$(d) P_T = P_p^\circ x_p + P_h^\circ x_h$$

$$= 440 \times \frac{1}{5} + 120 \times \frac{4}{5}$$

$$= 88 + 96 = 184;$$

$$P_p^\circ x_p = y_p P_T;$$

$$\frac{88}{184} = y_p$$

$$y_p = 0.478$$

81. Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R

Assertion A :- Carbon forms two important oxides –CO and CO_2 . CO is neutral whereas CO_2 is acidic in nature.

Reason R :- CO_2 can combine with water in a limited way to form carbonic acid, while CO is sparingly soluble in water.

In the light of the above statements, choose the most appropriate answer from the options given below :-

- A) Both A and R are correct but R is NOT the correct explanation of A.
B) Both A and R are correct and R is the correct explanation of A.
C) A is not correct but R is correct.
D) A is correct but R is not correct.

Solution : (Correct Answer: B)

The oxide which form acid on dissolving in water is acidic oxide.

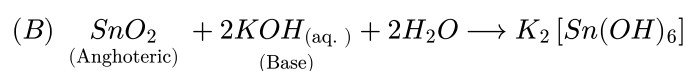
82. Choose the correct statement(s) among the following.

- (A) $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ is a reducing agent.
(B) SnO_2 reacts with KOH to form $\text{K}_2[\text{Sn}(\text{OH})_6]$.
(C) A solution of PbCl_2 in HCl contains Pb^{2+} and Cl^- ions.
(D) The reaction of Pb_3O_4 with hot dilute nitric acid to give PbO_2 is a redox reaction.

- A) A, C B) A, D C) A, B D) A, B, C

Solution : (Correct Answer: C)

(A) $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ is a reducing agent since Sn^{2+} tends to convert into Sn^{4+} .

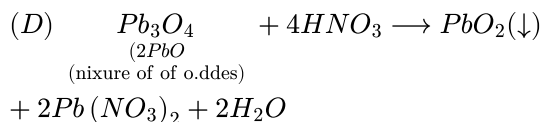


(C) First group cations (Pb^{2+}) form insoluble chloride with HCl that is PbCl_2 however it is slightly soluble in water and therefore lead +2 ion is never completely precipitated on adding hydrochloric acid in test sample of Pb^{2+} , rest of the Pb^{2+} ions are quantitatively precipitated with H_2S in acidic medium.

So that we can say that filtrate of first group contain solution of PbCl_2 in HCl which contains Pb^{2+} and Cl^-

However in the presence of conc. HCl or excess HCl it can produce $\text{H}_2[\text{PbCl}_4]$

So, we can conclude A, B or A,B,C should be answers.



It is not a redox reaction.

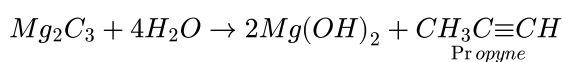
83. Select correct statement (s).

- A) Cyanamide ion (CN_2^{2-}) is isoelectronic with CO_2 and has the same linear structure
 B) Mg_2C_3 reacts with water to form propyne
 C) CaC_2 has $NaCl$ type lattice
 D) All of the above

Solution : (Correct Answer: D)

In CO_2 we have $22(6 + 8 + 8 = 22)$ electrons.

In (CN_2^{2-}) , we have $22(6 + 7 + 7 + 2 = 22)$ electrons. Both CO_2 and (CN_2^{2-}) have linear structures. Thus, statement (a) is correct.



i.e., statement (b) is also correct.

The structure of CaC_2 is of $NaCl$ type

i.e., statement (c) is also correct.

84. Which of the following compound formed silicones on hydrolysis

- A) $(CH_3)SiCl_2$ B) $(SiH_3)_3N$
 C) $SiCl_4$ D) All

Solution : (Correct Answer: A)

Silicones are a group of organosilicon polymers, which have $(-R_2SiO-)$ as a repeating unit. The starting materials for the manufacture of silicones are alkyl or aryl substituted silicon chlorides, $R_nSiCl_{(4-n)}$, where R is alkyl or aryl group. On hydrolysis of this alkyl or aryl substituted silicon chlorides the silicones are formed.

Among given options, option A is the only alkyl substituted silicone chloride and hence it is correct answer.

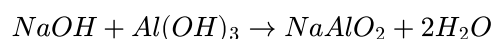
85. Assertion : $Al(OH)_3$ is insoluble in NH_4OH but soluble in $NaOH$.

Reason : $NaOH$ is strong alkali.

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 C) If the Assertion is correct but Reason is incorrect.
 D) If both the Assertion and Reason are incorrect.

Solution : (Correct Answer: A)

$Al(OH)_3$ is soluble in strong alkali like $NaOH$ because of formation of meta-aluminate ion

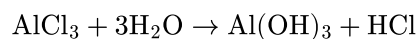


86. Heating an aqueous solution of aluminium chloride to dryness will give:

- A) $AlCl_3$ B) Al_2Cl_6
 C) Al_2O_3 D) $Al(OH)Cl_2$

Solution : (Correct Answer: C)

Aqueous solution of $AlCl_3$ is acidic as it gives HCL on hydrolysis.



On strongly heating $Al(OH)_3$ is converted into Al_2O_3



$AlCl_3$ on heating gives Al_2O_3

87. The solubility of anhydrous $AlCl_3$ and hydrous $AlCl_3$ in diethyl ether are S_1 and S_2 respectively. Then

- A) $S_1 = S_2$
 B) $S_1 > S_2$
 C) $S_1 < S_2$
 D) $S_1 < S_2$ but not $S_1 = S_2$

Solution : (Correct Answer: B)

Anhydrous $AlCl_3$ will dissolve the diethyl ether because anhydrous $AlCl_3$ is a good lewis acid.

It is electron deficient. The lone pairs of diethyl ether will be donated to the anhydrous $AlCl_3$ due to which it is soluble.

On the other hand, hydrous $AlCl_3$ have the water molecules which make it poor lewis acid.

So, $S_1 > S_2$

88. The species that do not contain peroxide linkage are :-

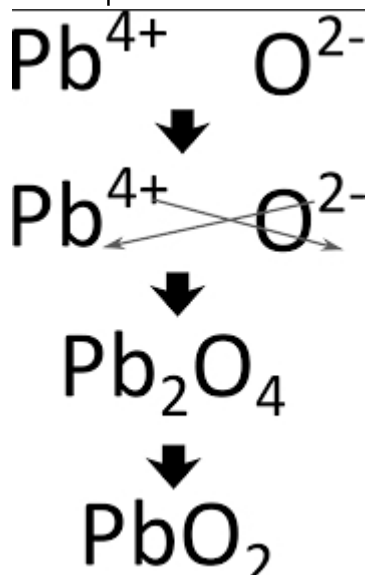
- A) PbO_2 B) H_2O_2
 C) SrO_2 D) BaO_2

Solution : (Correct Answer: A)

Metallic oxides which on treatment with dilute acids produce hydrogen peroxide are called peroxides. All peroxides contain a peroxide ion $(O_2)^{2-}$ having the structure $-O-O-$. PbO_2 does not contain a peroxide ion $(O_2)^{2-}$ and it can not be called as peroxides.

PbO_2 contains O^{2-} ions and Option (A) Pb is +4, whereas (B), (C) and (D) contains $[O-O]^{2-}$ ions (peroxide ions).

Thus Option A is correct.



89. Hoop's process is used for the purification of the metal

A) *Al* B) *Zn* C) *Ag* D) *Cu*

Solution : (Correct Answer: A)

(a) Hoop's process $\Rightarrow$ Purification of *Al*

Hall and Heroult process $\Rightarrow$ Reduction of Al_2O_3

Baeyer's and Serpeck's process $\Rightarrow$ Concentration of Bauxite ore

90. Alumina is

A) Acidic B) Basic
C) Amphoteric D) None of these

Solution : (Correct Answer: C)

(c) Alumina is amphoteric oxide, which reacts acid as well as base.

Biology - (Zoology)

91. Who proposed that the genetic code for amino acids should be made up of three nucleotides?

A) George Gamow B) Francis Crick
C) Jacques Monod D) Franklin Stahl

Solution : (Correct Answer: A)

George Gamow, a physicist proposed that genetic code for amino acids should be made up of three nucleotides.

92. Given below are two statements :

Statement I : Transfer RNAs and ribosomal RNA do not interact with mRNA.

Statement II : RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

In the light of the above statements, choose the most appropriate answer from the options given below :

- A) Both statement I and statement II are correct
B) Both statement I and statement II are incorrect
C) Statement I is correct but statement II is incorrect
D) Statement I is incorrect but statement II is correct

Solution : (Correct Answer: D)

Both transfer RNAs and ribosomal RNA interact with mRNA.

RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

93. Given below are two statements :

Statement I : In the RNA world, RNA is considered the first genetic material evolved to carry out essential life processes. RNA acts as a genetic material and also as a catalyst for some important biochemical reactions in living systems. Being reactive, RNA is unstable.

Statement II: DNA evolved from RNA and is a more stable genetic material. Its double helical strands being complementary, resist changes by evolving repairing mechanism.

In the light of the above statements, choose the most appropriate answer from the options given below :

- A) Both statement I and statement II are correct
B) Both statement I and statement II are incorrect
C) Statement I is correct but statement II is incorrect
D) Statement I is incorrect but statement II is correct

Solution : (Correct Answer: A)

In RNA world, RNA was the first genetic material as there are enough evidences to suggest that essential life processes (such as metabolism, translation, splicing, etc) evolved around RNA. RNA used to act as a genetic material as well as catalyst (there are some important biochemical reaction in living systems that are catalysed by RNA catalysts not by protein enzymes) so, statement I is correct statement II is also correct as DNA being double stranded and having complementary strands further resists changes by evolving a process of repair.

94. Which factor is important for termination of transcription?

- A) α (alpha) B) σ (sigma)
 C) ρ (rho) D) γ (gamma)

Solution : (Correct Answer: C)

In prokaryotes the RNA polymerase is only capable of catalysing the process of elongation. It associates transiently with initiation factor (σ) and termination factor (ρ) to initiate and terminate the transcription respectively.

95. Match List I with List II:

List - I	List - II
(a) Alfred Hershey and Martha Chase	(i) Streptococcus pneumoniae
(b) Euchromatin	(ii) Densely packed and dark-stained
(c) Frederick Griffith	(iii) Loosely packed and light-stained
(d) Heterochromatin	(iv) DNA as genetic material confirmation

Choose the correct answer from the options given below:

- A) A-II, B-IV, C-I, D-III
 B) A-IV, B-II, C-I, D-III
 C) A-IV, B-III, C-I, D-II
 D) A-III, B-II, C-IV, D-I

Solution : (Correct Answer: C)

The unequivocal proof that DNA is the genetic material came from the experiment of Alfred Hershey and Martha Chase.

Euchromatin are lightly stained region with loosely packed chromatin fibre.

Frederick Griffith performed series of experiments by selecting the different strains of Streptococcus pneumoniae.

Heterochromatin are darkly stained region with tightly packed chromatin fibre.

96. Which chromosome in the human genome has the highest number of genes?

- A) Chromosome X B) Chromosome Y
 C) Chromosome 1 D) Chromosome 10

Solution : (Correct Answer: C)

In human genome, Chromosome 1 has the highest number of genes, i.e., 2968.

97. Twins are born to a family that lives next door to you. The twins are a boy and a girl. Which of the following must be true?

- A) They are monozygotic twins.
 B) They are fraternal twins.
 C) They were conceived through in vitro fertilization.
 D) They have 75% identical genetic content.

Solution : (Correct Answer: B)

Fraternal twins or dizygotic twins are 2 separate fertilized eggs, they usually develop 2 separate amniotic sacs, placentas and supporting structures.

If twins are a boy and a girl, this indicates they are fraternal twins.

98. Histones are enriched with -

- A) Lysine & Arginine
 B) Leucine & Lysine
 C) Phenylalanine & Leucine
 D) Phenylalanine & Arginine

Solution : (Correct Answer: A)

In eukaryotes, packaging of DNA is much more complex. There is a set of positively charged, basic proteins called histones.

Histones are organised to form a unit of light molecules called histone octamer.

They are rich in the basic amino acid residues lysine and arginine.

99. Which of the following are the post-transcriptional events in an eukaryotic cell?

- A. Transport of pre-mRNA to cytoplasm prior to splicing.
 B. Removal of introns and joining of exons.
 C. Addition of methyl group at 5' end of hnRNA.
 D. Addition of adenine residues at 3' end of hnRNA.
 E. Base pairing of two complementary RNAs.

Choose the correct answer from the options given below :

- A) A, B, C only B) B, C, D only
 C) B, C, E only D) C, D, E only

Solution : (Correct Answer: B)

The process of copying genetic information from one strand of the DNA into RNA is known as transcription. It occurs in the cytoplasm with the help of transcribing enzyme.

Transport of pre-mRNA to cytoplasm prior to splicing is a part of transcription.

The primary transcript is converted into functional mRNA after post transcriptional processing involves 3 steps as follows-

- Modification of 5' end by capping,
- Tailing,
- Splicing.

Base pairing of two complementary RNA is not an event of post-transcription. Hence, statements B, C, D are post-transcriptional modification events in eukaryotic cell.

100. Given below are some stages of human evolution.

Arrange them in correct sequence. (Past to Recent)

A. Homo habilis B. Homo sapiens

C. Homo neanderthalensis D. Homo erectus

Choose the correct sequence of human evolution from the options given below:

- A) B – A – D – C B) C – B – D – A
C) A – D – C – B D) D – A – C – B

Solution : (Correct Answer: C)

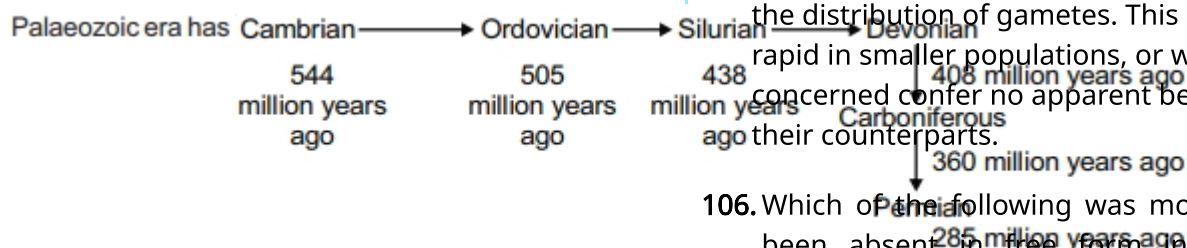
Correct answer is option (3) because the correct sequence of stages of human evolution from past to recent is

Homo habilis → Homo erectus → Homo neanderthalensis → Homo sapiens

101. The Permian period, during which the first most modern orders of insects appeared, occurred approximately

- A) 80 million years ago
B) 150 million years ago
C) 280 million years ago
D) 550 million years ago

Solution : (Correct Answer: C)



102. Which of the following is not the result of convergent evolution?

- A) Forelimbs of whales and bats
B) Flippers of penguins and dolphins
C) Potato and sweet potato
D) Eye of the octopus and mammals

Solution : (Correct Answer: A)

Both are mammals. They have similar structure of forelimbs and are a result of divergent evolution.

103. Which theory argues that life on earth came from outer space?

- A) Theory of panspermia B) Cosmozoic theory
C) Spore theory D) All of these

Solution : (Correct Answer: D)

Theory of panspermia (cosmozoic) was proposed by, Richter (1865). According to this theory, 'protoplasm' reached the earth in the form of spores of germs or other simple particles from some unknown parts of the universe with the cosmic dust, and subsequently evolved into various forms of life

104. What was not used by Millar in his experiment ?

- A) Condenser, electrodes, CO_2 , and O_2 .
B) CH_4 , H_2 , H_2O , NH_3 , boiling water.
C) Electrodes, boiling water, CH_4 , NH_3 .
D) Gases (CH_4 , NH_3 , H_2O , H_2), boiling water, condenser.

Solution : (Correct Answer: A)

105. Random unidirectional change in allele frequencies that occurs by chance in all populations and especially in small populations is known as

- A) migration B) natural selection
C) genetic drift D) mutation.

Solution : (Correct Answer: C)

(c) : Genetic drift (Sewall Wright effect) is the random change in the frequency of alleles in a population over successive generations in the gametes. Each new generation differs from its parental generation with regard to allele frequencies simply because of random variation in the distribution of gametes. This process is more rapid in smaller populations, or when the alleles concerned confer no apparent benefit compared to their counterparts.

106. Which of the following was most likely to have been absent in free form in the primordial atmosphere at the time of origin of life

- A) O_2 B) CH_4
C) H_2 D) NH_3

Solution : (Correct Answer: A)

It's Obvious

107. Contribution in chemical evolution in origin of life

A) Miller **B)** Darwin **C)** Lamarck **D)** Wallace

Solution : (Correct Answer: A)

(a) Miller and Urey performed an experiment to explain the chemical evolution in the origin of life.

108. According to one of the most widely accepted theories the earth's atmosphere, before any life had originated on this planet, consisted of a mixture of

- A)** Water vapours, H_2 , NH_3 and CH_4
B) H_2 , CO_2 , NH_3 and CH_4
C) O_2 , NH_3 , CH_4 and water vapours
D) Ozone, CH_4 , O_2 and water vapours

Solution : (Correct Answer: A)

It's Obvious

109. Given below are two statements:

Statement I:

Mendel studied seven pairs of contrasting traits in pea plants and proposed the Laws of Inheritance

Statement II :

Seven characters examined by Mendel in his experiment on pea plants were seed shape and colour, flower colour, pod shape and colour, flower position and stem height

In the light of the above statements, choose the correct answer from the options given below:

- A)** Both Statement I and Statement II are incorrect
B) Statement I is correct but Statement II is incorrect
C) Statement I is incorrect but Statement II is correct
D) Both Statement I and Statement II are correct

Solution : (Correct Answer: D)

110. Mendel conducted experiments for

- A)** 7 years **B)** 6 years
C) 5 years **D)** 4 years

Solution : (Correct Answer: A)

Mendel carried out hybridization experiments on garden pea for 7 years from 1856 – 1863

111. A son of a colourblind father and normal homozygous mother marries to a daughter of normal father and heterozygous mother than their children will be

- A)** All colourblind sons
B) All colourblind sons and daughters
C) Some colourblind and some normal daughters
D) Some normal and some colourblind sons

Solution : (Correct Answer: D)

(d) The son of colourblind man and homozygous normal mother will be normal and when he marries the daughter of a carrier mother which is expected to be carrier.

The progeny of such a couple will produce 50% colourblind sons and 50% normal sons.

112. Which of the following has been used for genetic researches

- A)** Pisum **B)** Neurospora
C) E. coli **D)** All the above

Solution : (Correct Answer: D)

It's Obvious

113. Alleles are form of

- A)** Genes **B)** Chromosomes **C)** DNA **D)** None

Solution : (Correct Answer: A)

(d) Allele is an alternative form of a gene for example, T and t alleles represent tall and short stem height of pea plant.

114. The colour of flower in pea is always linked with

- A)** Cotyledon **B)** Pod **C)** Leaves **D)** Seed coat

Solution : (Correct Answer: D)

(d) The colour of flower in pea is always linked with seed coat. Dominant trait is coloured and recessive trait is white.

115. A 1 : 1 phenotypic ratio in a test cross indicates that the

- A)** Alleles are codominant
B) Dominant phenotype of parent was heterozygous
C) Alleles segregated independently
D) Alleles are dominant

Solution : (Correct Answer: B)

It's Obvious

116. Mendelism is related with

- A)** Heredity in living beings
B) Meiosis during sexual reproduction
C) Mutation in living organisms
D) None of the above

Solution : (Correct Answer: A)

(a) Gregor Johann Mendel was the first to formulate clear cut laws of heredity.

117. The branch of botany dealing with heredity and variation is called

- A)** Geobotany **B)** Sericulture

- C)** Genetics **D)** Evolution

Solution : (Correct Answer: C)

(c) Genetics is the study of principles and mechanism of heredity and variations.

118. Which of the following techniques are used in IVF?

(I.) ZIFT (II), IUT (III) ICSI(IV.) AI(V.) IUI

- A)** II and III **B)** I and III
C) I and II **D)** II and III

Solution : (Correct Answer: D)

In IVF (In vitro fertilisation) the technique used are

ZIFT—Zygote Intra Fallopian Transfer

IUT— Intra Uterine Transfer

GIFT—Gamete Intra Fallopian Transfer

ICSI— Intra Cytoplasmic Sperm Injection

AI—Artificial Insemination

IUI— Intra Uterine Insemination

119. Which of the following is not included under barrier methods of birth control?

- A)** Vaginal pouch **B)** Diaphragm
C) Cervical cap **D)** Implant

Solution : (Correct Answer: D)

Implants are hormone containing devices which are implanted subdermally for providing long term contraception. e.g., norplant and implanon.

120. Periodic abstinence is avoiding sex during

- A)** Luteal phase **B)** Ovulatory phase
C) Menstrual phase **D)** None of these

Solution : (Correct Answer: B)

Avoid sex in ovulatory phase. Probable ova releasing day is 12 – 14 day of menstrual phase, and sperm can live for 2 – 3 days in female reproductive system. So, 10 – 17 days are the most probable days for fertilisation. Draw backs of periodic abstinence The effectiveness of this method is limited because only a few women have regular menstrual cycle and the actual time of ovulation can not be produced as the ovulation in humans occurs about 14 days before the onset of the next menstruation

121. Identify the correct statements

- i. Birth control pills are likely to cause cardiovascular problem
ii. A woman who substitutes or takes the place of the real mother to nurse to embryo is called surrogate mother

- iii. Numerous children have been produced by in vitro fertilisation but with some abnormalities
iv. Woman plays a key role in the continuity of the family and human species
v. Foetal sex determination test should not be banned

- A)** I and II
B) II and IV
C) III and V
D) I, II and IV

Solution : (Correct Answer: D)

Oral contraceptive pills increases the risk of intra vascular clotting. Therefore, they are not recommended for women with a history of disorders of blood clotting, careful blood vessel damage, hypertension, liver malfunction, heart disease or cancer of the breast or reproductive system

122. The permissible use of the technique amniocentesis is for

- A)** Detecting any genetic abnormality
B) Detecting sex of the unborn foetus
C) Artificial insemination
D) Transfer for embryo into the uterus of a surrogate mother

Solution : (Correct Answer: A)

Amniocentesis It is technique of drawing-amniotic fluid and testing it to find out the sex and disorders of the foetus. The amniotic fluid is usually taken at about the fourteenth or fifteenth week after conception.

Sonography (a technique involving high frequency sound waves to locate tissues of different densities within the body) is used to determine the location of the foetus and the placenta just prior to perform amniocentesis. Therefore, accidental damage to the foetus and placenta is prevented.

Amniotic fluid contains cell from the skin of the foetus and other sources. These cells can be used to determine the sex of the infant, to identify some abnormalities in the number of chromosomes and to detect certain biochemicals and enzymatic abnormalities.

If it is established that, if the child is likely to suffer from a serious increase congenital defect, the mother should get the foetus aborted. However, these days, this technique is also being misused. Since, the sex of the foetus can be determined by

this technique, normal female foetus is aborted. Therefore, this technique has been banned

Population growth rate is the annual average growth rate. It can be negative.

Birth rate replacement level can never be negative

123. Action of vaginal diaphragm is

- A) Prevent the ova to come in the uterus
- B) Prevent the sperm to come in contact with ova**
- C) Spermicidal
- D) Anti-implantational

Solution : (Correct Answer: B)

(b) Diaphragm and cervical cap are dom-shaped rubber plastic covers that are fitted on the cervix in the females vagina and check the entry of sperms into the uterus.

124. Amniocentesis is the withdrawal of amniotic fluid in

- A) Menopause
- B) Lactation
- C) Gestation
- D) Pregnancy**

Solution : (Correct Answer: D)

(d) Amniotic fluid is secreted by amnion of foetus during pregnancy.

125. The success of birth control programmes in controlling population growth is dependent on

- A) Use of contraceptives
- B) Tubectomy
- C) Vasectomy
- D) Acceptability of the above by the people**

Solution : (Correct Answer: D)

It's Obvious

126. Population of India in early 21st century may be

- A) 105 crore**
- B) 125 crore
- C) 95 crore
- D) 155 crore

Solution : (Correct Answer: A)

It's Obvious

127. Which gland's secretion is act as lubricants in male?

- A) Bulbourethral gland**
- B) Pancrease
- C) Seminal vesicle
- D) Vas deferens

Solution : (Correct Answer: A)

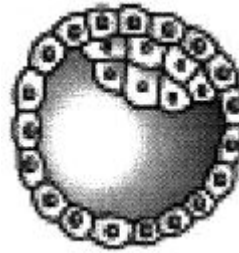
128. Fertilization takes place in

- A) Cervix
- B) Isthmus
- C) Ampulla**
- D) Follicle

Solution : (Correct Answer: C)

Fertilization takes place in ampulla of oviduct or ampullary isthmic junction

129. Identify figure



- A) Morula
- B) Blastocyst**
- C) Trophoblast
- D) Blastomere

Solution : (Correct Answer: B)

130. The organ which produces gametes are called ...A... and which neither produces gametes nor hormones are called ...B.... Here A and B represent

- A) A—primary sex organs; B—secondary sex organs**
- B) A— secondary sex organs; B— primary sex organs
- C) A—tertiary sex organs; B—secondary sex organs
- D) A— secondary sex organs; B— tertiary sex organs

Solution : (Correct Answer: A)

A—Primary sex organs; B—Secondary sex organs

131. Seminal plasma in human males is rich in

- A) Fructose and calcium**
- B) Glucose and Calcium
- C) DNA and testosterone
- D) Ribose and potassium

Solution : (Correct Answer: A)

Seminal plasma is composed of the fluid and sperms from the vas deferens (about 10% of the total), fluid from the seminal vesicles (almost 60%), fluid from the prostate gland (about 30%) and small amount of mucous gland secretions, especially the bulbourethral glands secretions. It contains calcium, citrate ion, phosphate ion a clotting enzyme, profibrinolysin, fructose, citrate, inositol, prostaglandins, several proteins, etc.

132. Choose incorrect pair.

- A) Interstitial cells : male sex hormone testosterone
- B) Sertoli cells : provide nutrition to the developing sperms
- C) Seminal vesicle : produce seminal plasma
- D) Prostate gland : responsible for sperm formation**

Solution : (Correct Answer: D)

133. Human uterus is

- A) Paired with well separated oviducts
- B) Single large chamber with posterior part of oviduct fused to it anteriorly**
- C) Paired with partially fused oviducts
- D) A single large chamber with completely fused oviducts

Solution : (Correct Answer: B)

It's Obvious

134. Testis produce

- A) Sperms** B) Eggs C) Seeds D) Spores

Solution : (Correct Answer: A)

It's Obvious

135. The rupture of the graafian follicle and the release of ovum occurs under the influence of

- A) LH** B) FSH C) MSH D) GH

Solution : (Correct Answer: A)

(a) Ovulation occurs under the influence of *LH* from pituitary gland.

Biology - (Botany)

136. Which of the following *RNAs* is not required for the synthesis of protein?

- A) mRNA B) tRNA C) rRNA **D) siRNA**

Solution : (Correct Answer: D)

137. *DNA* polymorphism forms the basis of :

- A) *DNA* finger printing
- B) Both genetic mapping and *DNA* finger printing**
- C) Translation
- D) Genetic mapping

Solution : (Correct Answer: B)

138. Which one of the following pair is a purine pair?

- A) Uracil, Guanine B) Cytosine, Thymine
- C) Adenine, Guanine** D) Adenine, Thymine

Solution : (Correct Answer: C)

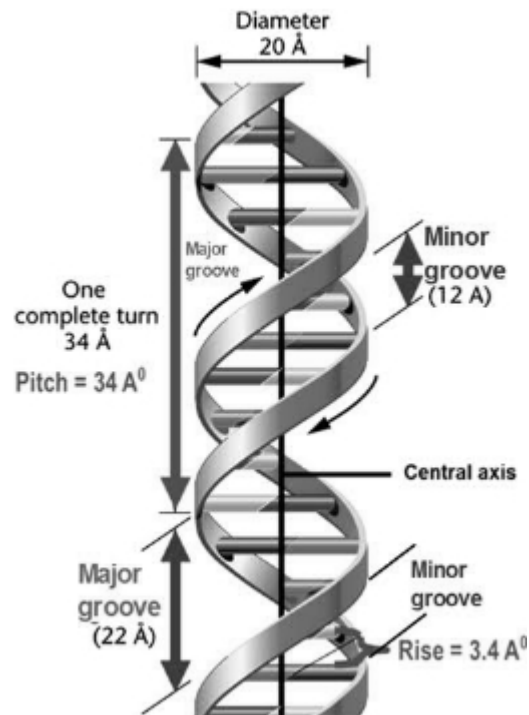
DNA (deoxyribose nucleic acid) consists of 3 different molecules-phosphate, 5 carbon deoxyribose sugar and nitrogenous base. The nitrogenous base may be a 9 - membered, double purine, *i.e.*, adenine (*A*) or guanine (*G*), or a 6 - membered, single -ringed pyrimidine, *i.e.*, thymine (*T*) or cytosine (*C*).

139. Which of the following is correct for Watson and Crick's model of *DNA*. It is duplex with

- A) 10 base pairs and 34 Å distance for every turn.**
- B) 10 base pairs and 3.4 Å distance for each turn of spiral.
- C) 20 base pairs and 34 Å for each turn.
- D) None of the above

Solution : (Correct Answer: A)

According to Watson and Crick model of *DNA*, it is a double helical molecule with 10 base pairs and 3.4 nm 34 Å distance for every turn.



140. Assertion : In a *DNA* molecule, *A – T* rich parts melt before *G – C* rich parts.

Reason : In between *A* and *T* there are three *H–* bond, whereas in between *G* and *C* there are two *H–* bonds.

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- C) If the Assertion is correct but Reason is incorrect.**
- D) If both the Assertion and Reason are incorrect.

Solution : (Correct Answer: C)

In a *DNA* molecule, *A – T* rich parts melt before *G – C* rich parts because there are two *H–* bond between *A* and *T* whereas in between *G* and *C*, there are three *H–* bond.

141. Purines found both in *DNA* and *RNA* are

- A) Adenine and thymine
- B) Adenine and guanine**

- C) Guanine and cytosine
D) Cytosine and thymine

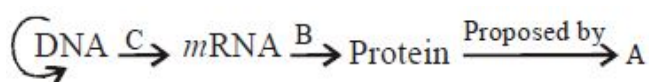
Solution : (Correct Answer: B)

142. Under which of the following conditions will there be no change in the reading frame of following *mRNA* 5' AACAGCGGUGCUAUU3'

- A) Insertion of G at 5th position
B) Deletion of G from 5th position
C) Insertion of A at G at 4th and 5th positions respectively
D) Deletion of GGU from 7th, 8th and 9th positions

Solution : (Correct Answer: D)

143. The figure gives an important concept in the genetic implication of *DNA*. Fill the blanks A, B and C.



- A) A-Maurice Wilkins, B-Transcription, C-- Translation
B) A-James Watson, B-Replication, C-Extension
C) A-ErwinChargaff, B-Translation, C-Replication
D) A-FrancisCrick, B-Translation, C-Transcription

Solution : (Correct Answer: D)

(d) : The expression of the genetic material occurs normally through the production of proteins. This involves two consecutive steps. These are transcription and translation. The *DNA* codes for the production of messenger *RNA* (*mRNA*) during transcription. Messenger *RNA* carries coded information to ribosomes. The ribosomes read this information and use it for protein synthesis. This process is called translation. *F.H.C.* Crick described this unidirectional flow of information in 1958 as the 'central dogma of molecular biology'.

144. Octamer which form complex unit with *DNA* is called

- A) Nucleosome B) Centrosome
C) Chromosome D) Endosome

Solution : (Correct Answer: A)

It's Obvious

145. Genetically active area of chromosome is called

- A) Euchromatin B) Heterochroatin
C) Heptan D) Cistron

Solution : (Correct Answer: A)

(a) Euchromatin or dynamic chromatin is relatively expanded and open. It has the potential of being actively transcribed.

146. The modern concept of gene is

- A) A segment of *DNA* capable of crossing over
B) A functional unit of *DNA*
C) A segment of *DNA*
D) A segment of chromosome

Solution : (Correct Answer: B)

(b) A gene is a segment of *DNA* which contains the information for one enzyme or one polypeptide chain coded in the language of nitrogenous bases or the nucleotides.

147. Who used heavy nitrogen N^{15} for proving semiconservative way of *DNA* replication

- A) Masters and Broda B) Meselson and Stahl
C) Watson and Crick D) Jacob and Monad

Solution : (Correct Answer: B)

(b) Meselson and stahl (1958) cultured *E. coli* bacteria in a culture medium to prove that *DNA* replication is semi conservative.

148. Adenine is 30% then what would be the % of guanine

- A) 10% B) 20% C) 30% D) 40%

Solution : (Correct Answer: B)

(b) The percentage of A + G equals 50% so percentage of guanine is 20%.

149. A nucleoside differs from a nucleotide in not having

- A) Phosphate B) Sugar
C) Nitrogen base D) Phosphate and sugar

Solution : (Correct Answer: A)

(a) In a nucleotide, purine or pyrimide nitrogenous base is joined by deoxyribose pentose sugar (d), which is further linked with phosphate (P) group to form nucleotides.

150. Wild type *E. coli* cells are growing in normal medium with glucose. They are transferred to a medium containing only lactose as the sugar. Which one of the following changes take place

- A) The lac-Operon is repressed
B) All Operons are induced
C) *E. coli* cells stop dividing
D) The lac-Operon is induced

Solution : (Correct Answer: D)

(d) Lac operon : cluster of genes that regulates production of enzymes needed to metabolize lactose in bacterial cells (*E. coli*). It is an inducible operon.

151. 'Central Dogma' was proposed by

- A) Crick
- B) Beadle and Tatum
- C) Temin and Baltimore
- D) Klug

Solution : (Correct Answer: A)

(a) The concept of central dogma was proposed by crick in 1958.

152. Transcription is

- A) Formation of *RNA* from *DNA*
- B) Formation of *DNA* from *RNA*
- C) Formation of protein
- D) Transfer of *RNA*

Solution : (Correct Answer: A)

It's Obvious

153. Which of the following is nongenetic, which is utilised for protein synthesis

- A) *DNA*
- B) *zRNA*
- C) *mRNA*
- D) None of these

Solution : (Correct Answer: C)

(c)The *mRNA* carries the message from *DNA* about the sequence of particular amino acids to be joined to form a polypeptide hence it is necessary for protein synthesis.

154. Which one of the following is common to both prokaryotes and eucaryotes

- A) Mitotic apparatus
- B) Histones
- C) Mitochondria
- D) Genetic code

Solution : (Correct Answer: D)

(d)The same genetic code is said to be present in all kinds of living organisms including virus bacteria, unicellular and multicellular organisms.

155. What is meant by sense strand

- A) The strand of *mRNA* involved in protein synthesis
- B) The strand of *tRNA* which starts protein synthesis
- C) The *DNA* strand which acts as a template for *mRNA* transcription
- D) Gradually growing molecule of protein

Solution : (Correct Answer: C)

(c)Sense strand also called coding strand or template strand.

156. Isolation and purification of specific *DNA* segment from a living organism was achieved by

- A) Crick
- B) Nirenberg
- C) Khorana
- D) Beckwith and his colleagues

Solution : (Correct Answer: B)

It's Obvious

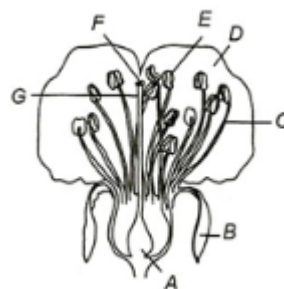
157. The strongest evidence that *DNA* is genetic material comes from

- A) That chromosomes are made up of *DNA*
- B) Transformation of bacterial cells
- C) That *DNA* is not present in the cytoplasm
- D) That *DNA* is present in the nucleus

Solution : (Correct Answer: B)

(b) Transformation of bacterial cell by pneumonia bacteria and rat experiment.

158. Identify *A* to *G* in following figure and answer accordingly



- A) *A*—Ovary, *B*—Filament, *C*—Sepal, *D*—Petal, *E*—Style, *F*—Stigma, *G*—Anther
- B) *A*—Petal, *B*—Ovary, *C*—Petal, *D*—Filament, *E*—Anther, *F*—Stigma, *G*—Style
- C) *A*—Ovary, *B*—Sepal, *C*—Filament, *D*—Petal, *E*—Anther, *F*—Stigma, *G*—Style
- D) *A*—Ovary, *B*—Sepal, *C*—Filament, *D*—Petal, *E*—Anther, *F*—Stigma, *G*—Style

Solution : (Correct Answer: C)

A

159. Which of the following is not a characteristic feature of insect pollinated flowers?

- A) Fragrance
- B) Nectaries
- C) Foul odour
- D) Mucilaginous covering on pollen grains

Solution : (Correct Answer: D)

Mucilage covering in pollen grain is found in aquatic plants.

160. The process of formation of microspores from a pollen mother cell is called

- A) Megasporogenesis
- B) Microsporogenesis
- C) Megagametogenesis
- D) Microgametogenesis

Solution : (Correct Answer: B)

MMC (Microspore mother cell) Meiosis Haploid Microspore

161. Which one of the following would not lead to formation of clones?

- A) Double fertilization B) Apomixis
C) Vegetative reproduction D) Tissue culture

Solution : (Correct Answer: A)

Clone is an individual obtaining from single parent through apomixis, vegetative reproduction and tissue culture. The process of fusion of two male gametes in a single embryo sac is called double fertilization. It is found in sexual reproduction of angiosperms only and discovered by Nawaschin (1898).

162. Sexual reproduction leads to

- A) Genetic recombination B) Polyploidy
C) Aneuploidy D) euploidy

Solution : (Correct Answer: A)

Sexual reproduction includes syngamy and meiosis. Syngamy is the nuclear fusion of male and female gamete. The meiosis is reduction of chromosome number to haploid during meiosis. Genetic recombination occurred as a result of crossing over.

163. Flower is a

- A) Modified male plant only
B) Modified female plant only
C) Modified reproductive shoot
D) Vegetative shoot system

Solution : (Correct Answer: D)

Flower is a modified shoot meant for reproduction

164. In flowering plants meiosis occurs at the time of

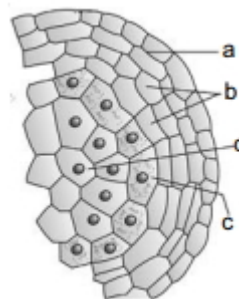
- A) Formation of buds
B) Germination of seed
C) Formation of root primordia
D) Formation of pollen grains

Solution : (Correct Answer: D)

In flowering plants, a cross section of the developing anther displays four chambers. These chambers are called pollen sacs. Each pollen sac is filled with cells containing large nuclei. As the anther grows each of these cells goes through two meiotic divisions, forming a tetrad. These cells are called microspores. Each one of these microspores eventually becomes a pollen grain and in carpel meiosis takes place at the time of megaspore from megaspore mother cell

165. Examine the figure given below and select the right option giving all the four parts *a, b, c* and *d*. Correctly identify

a b c d



- A) Endothecium Tapetum Microspore mother cell Middle layers
B) Tapetum Endothecium Microspore mother cell Middle layers
C) Endothecium Middle layer Tapetum Microspore mother cell
D) Endothecium Microspore mother cell Middle layer Tapetum

Solution : (Correct Answer: C)

166. Which of the following statement is applicable for all flowering plants?

- A) Monosiphonous pollen tube
B) Non-motile and morphologically dissimilar gametes
C) Presence of pollinium
D) Division of generative cell after pollination

Solution : (Correct Answer: B)

Pollen tube will carry the gametes.

167. Seed germination requires

- I. Light II. Temp (suitable)*
III. Moisture IV. Oxygen
Select correct option

- A) *I, II and III*
B) *II, III and IV*
C) *I, III and IV*
D) *II, IV and I*

Solution : (Correct Answer: B)

During the germination the light is not needed. But later stage of development light plays a greater role in making food

168. What does the filiform apparatus do at the entrance into ovule?

- A) It brings about opening of the pollen tube.
B) It guides pollen tube from a synergid to egg.
C) It helps in the entry of pollen tube into a synergid.

- D) It prevents entry of more than one pollen tube into the embryo sac.

Solution : (Correct Answer: B)

(b) : Within the embryo sac three cells are grouped together at the micropylar end and constitute the egg apparatus. The egg apparatus, in turn, consists of two synergids and one egg cell. The synergids have special cellular thickenings at the micropylar tip called filiform apparatus, which plays an important role in guiding the pollen tubes into the synergid. Three cells are at the chalazal end and are called the antipodals. The large central cell, has two polar nuclei.

169. Diameter of pollen grain

- A) 25 – 50 $\text{\AA}$
- B) 25 – 50 micron
- C) 20 – 25 micrometre
- D) 25 – 50 micrometre

Solution : (Correct Answer: D)

170. Assertion : Endosperm is a nutritive tissue and it is triploid.

Reason: Endosperm is formed by fusion of secondary nucleus to second male gamete. It is used by developing embryo

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- C) If the Assertion is correct but Reason is incorrect.
- D) If both the Assertion and Reason are incorrect.

Solution : (Correct Answer: A)

Male gamete (n) + secondary nucleus ($2n$) = primary endosperm nucleus which develops into endosperm ($3n$) Endosperm is the reserve food used by developing embryo

171. A- Four cells are at the chalazal end and are called the antipodals.

R – A typical angiosperm embryo sac, at maturity, though 8-nucleate is 7-celled

- A) *A* – *R* correct
- B) *A* – *R* incorrect
- C) *A* -correct, *R* -incorrect
- D) *A* -incorrect, *R* -correct

Solution : (Correct Answer: D)

172. Because of it pollen grains are well preserved as fossils

- A) Presence of germ pore
- B) Presence of sporopollenin
- C) Presence of cellulose
- D) All correct

Solution : (Correct Answer: B)

173. Parthenocarpic tomato fruits can be produced by

- A) treating the plants with phenylmercuric acetate
- B) removing androecium of flowers before pollen grains are released
- C) treating the plants with low concentrations of gibberellic acid and auxins
- D) raising the plants from vernalized seeds.

Solution : (Correct Answer: C)

(c) : Development of fruits without fertilization is called parthenocarpy and such fruits are called parthenocarpic fruits. Parthenocarpic fruits are seedless. A flower is emasculated and auxins are applied to the stigma of the flower, it forms a parthenocarpic fruit. For parthenocarpy induction by auxins, these should be applied after anthesis fertilization, is called amphimixis. But in some plants, this normal sexual reproduction (amphimixis) is replaced by some abnormal type of sexual reproduction called apomixis. Apomixis may be defined as, 'abnormal kind of sexual reproduction in which egg or other cells associated with egg (synergids, antipodals, etc.) develop into embryo without fertilization and with or without meiosis'. Adventive embryony is a type of apomixis in which development of embryos directly takes place from sporophytic tissues like nucellus and integuments, *e.g.*, Citrus, mango, etc.

174. Which of the following statements is correct?

- A) Sporopollenin can be degraded by enzymes.
- B) Sporopollenin is made up of inorganic materials.
- C) Sporopollenin can withstand high temperatures as well as strong acids and alkalis.
- D) Sporopollenin can withstand high temperatures but not strong acids.

Solution : (Correct Answer: C)

(c) : Pollen grain is a haploid, unicellular body. It is cuticularised and the cutin is of special type called sporopollenin, which is resistant to chemical and biological decomposition. It can withstand high temperatures as well as strong acids and alkalis. This is why, pollen wall is preserved for long periods in fossil deposits. In addition pollen wall possesses proteins for enzymatic and compatibility reactions.

175. Pollination by water is seen in

- A) Nelumbium
- B) Vallisneria
- C) Eichornia
- D) Nymphaea

Solution : (Correct Answer: B)

(b)Vallisneria is sub merged, dioecious, fresh water plant.

176. Egg apparatus of angiosperm consists of

- A) One egg cell and two synergids
- B) One egg cell 2 synergids 3 antipodals
- C) 3 antipodals only
- D) Secondary nucleus and egg cell

Solution : (Correct Answer: A)

(a)Egg apparatus of angiosperm is 3 celled. It has one egg cell and two synergids on either side of egg cell. 3 antipodals are present on chalazal side. The secondary nucleus is binucleate is present in the centre of embryo sac.

177. Mature embryo sac contains or A normal angiosperm embryo sac at the final stage of development has

- A) 4 cells
- B) 3 cells
- C) 7 cells
- D) 8 cells

Solution : (Correct Answer: C)

(c)In mature embryo sac cells are always seven but nuclei are always eight.

178. Fig is pollinated by

- A) Wind
- B) Self
- C) Water
- D) Insects
(Blastophaga grossorum)

Solution : (Correct Answer: D)

It's obvious.

179. In a pollen grain, larger nucleus is

- A) Generative nucleus
- B) Tube nucleus
- C) Vegetative nucleus
- D) None of these

Solution : (Correct Answer: B)

(b)Tube has a large nucleus which is surrounded by thin sheath of cytoplasm.

180. The anther wall consists of four wall layers where

- A) Endothecium lies inner to middle layers
- B) Tapetum lies just inner to endothecium
- C) Tapetum lies next to epidermis
- D) Middle layers lie between endothecium and tapetum

Solution : (Correct Answer: D)

It's obvious.